BRIDGING COMMUNICATION GAPS FOR THE HEARING IMPAIRED

A PROJECT REPORT

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HEALTHIN RHOWINA ROCHE R

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DEPARTMENT OF INFORMATION SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING, GUINDY

ANNA UNIVERSITY

CHENNAI 600 025

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ANNA UNIVERSITY

CHENNAI - 600 025

BONAFIDE CERTIFICATE

Certified that this project report titled **"BRIDGING COMMUNICATION GAPS FOR THE HEARING IMPAIRED"** is the bona fide work of **HEALTHIN RHOWINA ROCHE R (2023179014)** who carried out project work under my supervision. Certified further that to the best of my knowledge and belief, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or an award was conferred on an earlier occasion on this or any other candidate.

PLACE:
DATE:

Dr. E. UMA
ASSOCIATE PROFESSOR
PROJECT GUIDE
DEPARTMENT OF IST, CEG
ANNA UNIVERSITY
CHENNAI 600025

COUNTERSIGNED

DR. S. SWAMYNATHAN
HEAD OF THE DEPARTMENT
DEPARTMENT OF INFORMATION SCIENCE AND TECHNOLOGY
COLLEGE OF ENGINEERING, GUINDY
ANNA UNIVERSITY
CHENNAI 600025

ABSTRACT

Real-time sign language recognition system designed to translate World Level Sign Language gestures into text and speech, encouraging communication between deaf and hearing individuals. Utilizing MediaPipe for hand landmark detection, the system extracts 294 spatial and motion based features per frame from webcam captured video sequences. A Random Forest classifier, trained on a dataset of 282 unique WLASL gestures, achieves an accuracy of 92% on a stratified test set. The preprocessing pipeline includes temporal augmentation, keypoint normalization, and motion feature computation to enhance gesture differentiation. An n-gram language model refines predictions into coherent sentences, leveraging contextual word transitions. The Tkinter-based graphical user interface provides live video feedback, gesture predictions, and multilingual audio output in English and Tamil using gTTS. The system processes 30 frame sequences, supports single and double hand gestures, and employs confidence based filtering to ensure reliability. Evaluation metrics, including accuracy, macro F1-score, and classification reports, demonstrate robust performance. This scalable, cost effective solution bridges communication gaps, promoting inclusivity for the deaf community.

திட்டப்பணிச்சுருக்கம்

உலகளாவிய புலப்படுத்தல் மொழி சைகைகளை உரையாகவும் பேச்சாகவும் மொழிபெயர்க்கும் நிகழ்நேர கையசைவு அறிதல் அமைப்பு, செவிடு மற்றும் கேட்கும் தனிநபர்களிடையே தொடர்பை ஊக்குவிக்கிறது. MediaPipe மூலம் கை முக்கிய புள்ளிகளை கண்டறிந்து, இந்த அமைப்பு ஒளிப்பதிவு கருவி மூலம் பதிவு செய்யப்பட்ட காட்சி வரிசைகளிலிருந்து ஒரு பிரேமுக்கு 294 இடஞ்சார்ந்த மற்றும் இயக்க அம்சங்களை பிரித்தெடுக்கிறது. 282 தனித்துவமான WLASL சைகைகளின் தரவுத்தொகுப்பில் பயிற்றுவிக்கப்பட்ட Random Forest classifier, பிரிக்கப்பட்ட சோதனைத் தொகுப்பில் 92% துல்லியத்தை அடைகிறது. முன்பதிவு செயலாக்கக் குழாய் நேர விரிவாக்கம், முக்கிய புள்ளி இயல்பாக்கம், மற்றும் இயக்க அம்ச கணிப்பு ஆகியவற்றை உள்ளடக்கி சைகை வேறுபாட்டை மேம்படுத்துகிறது. n-gram language model சூழல் சொல் மாற்றங்களை பயன்படுத்தி கணிப்புகளை ஒத்திசைவான வாக்கியங்களாக செம்மைப்படுத்துகிறது. Tkinter அடிப்படையிலான பயனர் இடைமுகம் நேரடி காட்சி கருத்து, சைகை கணிப்புகள், மற்றும் gTTS மூலம் ஆங்கிலம் மற்றும் தமிழில் பன்மொழி ஒலி வெளியீடு வழங்குகிறது. 30 பிரேம் வரிசைகளை செயலாக்கி, ஒற்றை மற்றும் இரட்டை கை சைகைகளை ஆதரிக்கும் இந்த அமைப்பு, நம்பிக்கை அடிப்படையிலான வடிகட்டல் மூலம் நம்பகத்தன்மையை உறுதிசெய்கிறது. துல்லியம், மேக்ரோ F1-score, மற்றும் வகைப்பாடு அறிக்கைகள் உள்ளிட்ட மதிப்பீட்டு அளவீடுகள் உறுதியான செயல்திறனை வெளிப்படுத்துகின்றன. இந்த அளவிடக்கூடிய, செலவு குறைந்த தீர்வு செவிடு சமூகத்திற்கு உள்ளடக்கத்தை ஊக்குவிக்கும் வகையில் தொடர்பு இடைவெளிகளை இணைக்கிறது.

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HEALTHIN RHOWINA ROCHE R

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LIST OF ABBREVIATIONS

3D	Three-Dimensional
AI	Artificial Intelligence
CV	Computer Vision
DL	Deep Learning
GTTS	Google Text-to-Speech
GUI	Graphical User Interface
ML	Machine Learning
RF	Random Forest
SLR	Sign Language Recognition
UI	User Interface
WLASL	Word-Level American Sign Language

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Sign language is a vital form of communication for millions of individuals with hearing and speech impairments worldwide. Through a complex combination of hand gestures and body movements, it allows the deaf and hard-of-hearing community to express their thoughts, emotions, and intentions with remarkable clarity and nuance. More than just a substitute for spoken language, sign language represents a rich and culturally significant mode of expression that fosters deep connection and mutual understanding within and beyond deaf communities.

However, despite its critical importance, a communication gap persists between those fluent in sign language and those who are not. Bridging this gap often necessitates the use of human interpreters or specialized assistive devices solutions that can be expensive, time-consuming, or inaccessible in many situations. The emergence of modern technologies, particularly in the fields of computer vision and machine learning, offers promising new avenues to address these challenges. Automated sign language recognition systems have the potential to revolutionize communication by translating gestures into spoken or written language in real time, thereby enabling more inclusive and seamless interactions.

This project presents a sophisticated system capable of recognizing and interpreting a broad vocabulary of 282 American Sign Language (ASL) gestures using video input from a standard webcam. Unlike earlier models that

depended on static images, this system emphasizes dynamic gesture recognition. It captures not only the spatial configuration of hand shapes but also the temporal movement patterns crucial for understanding full expressions and forming meaningful sentences.

The system leverages MediaPipe's advanced hand-tracking framework to accurately detect hand keypoints specific landmarks that define gesture shapes. These keypoints are then processed using a Random Forest classifier, a machine learning model known for its accuracy and reliability in pattern recognition tasks. This combination enables real-time gesture recognition, which is translated into both on-screen text and audio output in multiple languages, including English and Tamil.

What distinguishes this system is its seamless integration of advanced technology with a user-friendly design. A graphical user interface (GUI) developed using Tkinter ensures ease of use, offering real-time visual feedback, gesture predictions, and voice synthesis. This intuitive interface enhances the user experience and makes the system accessible to a wide range of users from individuals in need of assistive communication tools to educators and developers exploring sign language technologies.

The broader impact of this work goes beyond technological innovation. At its core, it strives to democratize communication for the deaf and hard-of-hearing community, enabling greater participation in social, educational, and professional settings. By using affordable, widely available hardware like standard webcams, and eliminating the need for specialized equipment or extensive training, the system provides a scalable and cost-effective alternative to traditional solutions.

Its potential applications are vast. In healthcare, the system could facilitate better communication between patients and medical professionals. In education, it could help create more inclusive classrooms where sign language users and non-signers can learn together. Furthermore, it opens possibilities in human-computer interaction, virtual reality, and language learning platforms.

Through careful development and evaluation, this project demonstrates the practicality and reliability of automated sign language recognition. It is a testament to the power of technology to break down communication barriers and promote inclusivity. By combining technological precision with a commitment to social impact, the system not only contributes to the field of assistive technology but also inspires further innovations aimed at creating a more accessible and equitable world for all.

1.2 PROBLEM STATEMENT

The challenge in sign language recognition lies in accurately interpreting a wide range of hand gestures, which vary in spatial configuration, motion patterns, and temporal dependencies. Gestures can be subtle, context-dependent and influenced by factors such as lighting, hand orientation, and individual signing styles. Key problems include extracting robust feature representations from video data that capture both static hand poses and dynamic movements, and training a machine learning model capable of classifying these features into a large vocabulary with high precision. Overcoming these challenges is essential for developing a practical system that supports real-time communication.

1.3 CHALLENGES

The development of an effective sign language recognition system involves overcoming several multifaceted challenges, each rooted in the complexity of gesture-based communication and the limitations of current technology.The challenges are as follows:

1.3.1 Variability in Gesture Execution

Sign language gestures vary between users due to differences in hand size, speed, and signing habits, necessitating a model that generalizes across various execution styles.

1.3.2 Limited Generalization

Designing a model that performs well on unseen gestures or environmental conditions such as lighting, background noise is complex, requiring robust feature extraction and adaptive learning.

1.3.3 Data Imbalance

Collecting a balanced data set with sufficient samples for all 282 gestures is difficult, as some signs may be underrepresented, potentially biasing the model.

1.3.4 Complexity of Feature Representation

Hand keypoints alone may not fully capture the nuances of dynamic gestures, requiring additional motion-based features to enhance accuracy.

1.3.5 Real Time Processing

Achieving low latency predictions in a live video feed poses computational challenges, critical for practical usability.

1.4 PROPOSED SOLUTION

The proposed solution involves a multi-stage pipeline for sign language recognition. Video data is processed using MediaPipe Hands to extract 3D hand keypoints, normalized relative to wrist position and hand size for consistency. These spatial features are augmented with motion velocity to capture temporal patterns. The resulting feature sequences are fed into a Random Forest classifier, trained on a dataset of 282 ASL gestures. Temporal augmentation and sentence formation logic, guided by n-gram probabilities, enhance prediction robustness and context awareness. A Tkinter-based interface provides real-time visualization, gesture classification, and multilingual audio output, ensuring accessibility and user engagement.

1.5 ORGANIZATION OF THE REPORT

Chapter 2 presents the Literature Review, discussing previous research studies, methodologies, and innovations in the field of sign language recognition, gesture translation.

Chapter 3 Explains the system design, including architecture and module details.

Chapter 4 Details the implementation, covering data collection, preprocessing, and UI development.

Chapter 5 Presents results and analysis, including model performance metrics and interface functionality.

Chapter 6 Summarizes the overall projection impact,highlights key achievement, and potential improvements

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

Sign Language Recognition (SLR) has solidified its place as a dynamic and transformative field, weaving together cutting-edge advancements from artificial intelligence (AI), computer vision, linguistics, and human computer interaction to create systems that bridge communication divides. At its heart, SLR seeks to empower individuals who are deaf, mute, or both by developing intelligent technologies capable of interpreting the intricate gestures of sign language and rendering them into spoken or written forms. This interdisciplinary endeavor goes beyond mere translation; it aims to foster seamless, natural interactions, enabling greater inclusion in social, educational, and professional settings. The convergence of AI's analytical prowess, computer vision's ability to parse visual data, linguistics' insights into language structure, and human-computer interaction's focus on intuitive design has propelled SLR into a realm of immense potential. By decoding the visual and kinetic elements of sign language hand shapes, movements, facial expressions, and body postures these systems strive to replicate the fluency and expressiveness of human communication, making accessibility a reality for millions worldwide.

The evolution of SLR reflects a remarkable journey from cumbersome, hardware reliant approaches to sophisticated, vision-based solutions that leverage the power of modern computing. Decades ago, early systems depended heavily on specialized equipment like data gloves and motion sensors, which, while effective in controlled settings, were often impractical due to their cost, complexity, and restrictive nature. These sensor-based methods

required users to wear devices that tracked hand movements, limiting their accessibility and comfort. In contrast, recent advancements have embraced markerless, vision based systems that rely on standard cameras, such as those found in webcams or smartphones, paired with deep learning techniques. This shift has democratized SLR, making it more scalable and user friendly. Deep learning models, trained on vast datasets of sign language videos, can now recognize complex gestures with increasing accuracy, adapting to the nuances of hand configurations and motion patterns. However, despite these strides, real time SLR systems face persistent challenges. Variations in how individuals perform signs, known as signer variability, can confuse models, as can gesture ambiguity, where similar movements convey different meanings. Environmental factors like fluctuating lighting, cluttered backgrounds, or occlusions where hands are partially obscured further complicate recognition. Scaling these systems to accommodate diverse sign languages and regional dialects remains a formidable hurdle, underscoring the need for continued innovation.

This chapter offers an in-depth exploration of the sign language recognition landscape, tracing its historical roots and spotlighting the latest breakthroughs that are shaping its future. It begins by chronicling the progression of SLR systems, from their hardware-heavy origins to the current era of vision based, AI-driven solutions, highlighting key technological milestones along the way. The survey then delves into recent advancements in gesture classification, examining how machine learning algorithms ranging from traditional models to deep neural networks have enhanced the precision of gesture recognition. Beyond recognition, the chapter explores techniques for integrating speech synthesis and animated avatars, which enrich the translation process by providing multimodal outputs that cater to diverse audiences. A comparative analysis of existing methodologies sheds light on their strengths and limitations, offering a balanced perspective on the state of the field. The

review culminates in a critical examination of research gaps, such as the need for more robust handling of environmental noise, better generalization across signers and languages, and improved real time performance. These gaps have directly inspired the design of the proposed system, which aims to address these challenges through innovative approaches, paving the way for more reliable, inclusive, and scalable SLR solutions that empower users and advance the broader mission of accessibility.

2.2 ENHANCING COMMUNICATION BETWEEN DEAF-MUTE INDIVIDUALS

Hanaa ZainEldin et al. (10) performed an extensive review of the applications of Artificial Intelligence (AI), Deep Learning (DL), and Machine Learning (ML) in facilitating communication for individuals who are deaf and mute. Their survey covers the integration of sign language interpretation, speech-to-text, and text-to-speech mechanisms across various platforms.

The review emphasizes the role of intelligent systems in enhancing accessibility and inclusion, highlighting the challenges involved in designing models capable of recognizing complex gestures in real-world conditions. Furthermore, their study identifies the need for adaptive and scalable models that can operate in multilingual and multicultural environments.

By consolidating existing approaches and highlighting emerging research directions, the work provides a valuable roadmap for future efforts aimed at bridging the communication gap through AI-driven solutions. The authors underscore the transformative potential of such technologies in improving the quality of life for the impaired community.

2.3 DEVELOPMENT OF A TWO-WAY SIGN LANGUAGE TRANSLATOR

Divya et al. (2) proposed a bi-directional communication system aimed at enabling real-time interaction between individuals with and without hearing impairments. Their system converts sign gestures into corresponding text or speech outputs, and vice versa, allowing for seamless two-way communication.

The architecture incorporates computer vision techniques to detect gestures and deep learning models for gesture classification and voice mapping. The system's novelty lies in its ability to dynamically translate voice input back into animated sign language, thus providing mutual understanding between the hearing and non-hearing communities.

2.4 GESTURE-BASED SIGN LANGUAGE SYSTEMS

Pulkit Singh et al. (8) developed a dual-mode translator that uses computer vision to interpret hand signs and neural networks to generate accurate text and speech. The system also leverages animation for the reverse conversion, enabling hearing individuals to interact with signers in a more intuitive and visual manner.

These innovations have significantly contributed to the development of closed-loop communication systems, eliminating the dependency on human interpreters and promoting autonomy in interpersonal exchanges.

Several researchers have focused on creating real-time gesture recognition systems capable of interpreting static and dynamic signs. Pulkit Singh et al. developed a prototype that records hand movements via a standard

camera and translates them into text and speech with an accuracy of 99.3%. The system emphasizes real-time interaction and low hardware dependency, making it viable for low-resource environments.

2.5 ADVANCED RECOGNITION WITH MACHINE LEARNING

Expanding on this idea, Naga Thulasi Vundelli et al.Introduced a more inclusive system that caters to individuals who are blind, deaf, and mute. Their approach integrates tactile feedback mechanisms alongside gesture detection and audio conversion to support users with multiple disabilities. The system not only bridges communication gaps but also promotes inclusivity through multi-modal interaction capabilities.

These gesture-based systems represent an important step toward fully autonomous communication tools. However, challenges persist in terms of gesture ambiguity, inter-user variability, and maintaining robustness across varying environmental conditions.

Pranali Kale et al. (4) introduced sign unite, an intelligent system for American Sign Language (ASL) detection. Utilizing machine learning algorithms, their system successfully recognizes and classifies a broad set of ASL gestures, facilitating real-time communication between the deaf-mute community and mainstream society.

Another noteworthy development involved the integration of MediaPipe for precise hand landmark extraction, combined with a lightweight Random Forest classifier to ensure rapid gesture classification. The system achieved a commendable accuracy of 94.69%, making it suitable for practical use in web-based applications. Notably, the system also supports bi-directional

communication among deaf-mute individuals, hard-of-hearing users, and non-signers.

These efforts demonstrate the feasibility of deploying machine learning models in real-time, browser-compatible environments. They also validate the effectiveness of combining landmark extraction with classical classification algorithms for robust gesture recognition.

2.6 CHALLENGES IN CROSS-LINGUISTIC SIGN LANGUAGE RECOGNITION

While much of the existing research focuses on American Sign Language (ASL), other sign languages, such as Indian Sign Language (ISL) or British Sign Language (BSL), exhibit unique grammatical structures and gesture vocabularies. A key challenge in cross-linguistic SLR is the lack of standardized datasets that capture these variations. For instance, Koller et al. (5) highlighted models trained on ASL datasets often fail to generalize to ISL due to differences in hand shapes and motion patterns. This limitation underscores the need for multilingual datasets and transfer learning techniques to adapt models across sign languages.

The proposed system, while focused on ASL, incorporates design principles that could facilitate adaptation to other sign languages. By using normalized keypoints and motion features, the system reduces dependency on language-specific hand configurations, paving the way for future extensions to languages like Tamil Sign Language. Such adaptability is crucial for global deployment, particularly in regions with diverse linguistic landscapes.

2.7 COMPARATIVE INSIGHTS AND OBSERVED LIMITATIONS

Upon reviewing the literature, several recurring trends and limitations become evident:

- Gesture Ambiguity: Many models struggle with distinguishing between visually similar gestures, especially those involving subtle finger or wrist movements.
- Signer Dependency: A significant number of systems are trained on limited datasets that do not generalize well across different signers or hand sizes.
- Environmental Sensitivity: Variations in lighting, background, and camera resolution often lead to degraded recognition accuracy.
- Limited Vocabulary: While some systems support a handful of static gestures, few have the capability to handle large vocabularies or continuous signing.
- Lack of Natural Feedback: Most systems lack avatar-based visual feedback, which can enhance comprehension for non-signers and improve educational applications.

2.8 SUMMARY OF THE LITERATURE REVIEW

The collective body of research highlights a transition from sensor-based to vision-based and AI-enhanced systems. Modern systems increasingly rely on Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and classical classifiers such as SVMs and Random

Forests to interpret complex gestures. While many of these systems achieve high accuracy in controlled environments, they fall short in practical deployment due to scalability issues and the absence of intuitive visual feedback.

To address these gaps, this project proposes a comprehensive hybrid system that leverages MediaPipe for keypoint extraction, motion-based feature engineering, Random Forest classification, and Tkinter-based GUI for output visualization. The system incorporates temporal dynamics (such as delta keypoints and velocity vectors),and supports multilingual audio output to deliver a seamless and accessible user experience.

This hybrid approach aims to overcome the limitations of previous works and set a new standard for inclusive, real-time sign language recognition technologies.

CHAPTER 3

SYSTEM DESIGN

The system design for this sign language recognition project presents a well structured and integrated framework that effectively transforms raw video input into meaningful linguistic output with precision and speed. This enables real-time communication for individuals who are deaf or hard of hearing. At its core, the system incorporates advanced computer vision techniques, sophisticated machine learning algorithms, and a user-friendly interface to create a seamless experience that bridges the communication gap between signers and non signers.

By utilizing a comprehensive vocabulary of 282 gestures from the Word Level American Sign Language (WLASL) dataset, the system achieves accurate gesture classification and coherent sentence construction. It captures both the subtle details of hand movements and their contextual relationships, enabling the expression of complete thoughts. This comprehensive approach positions the system not only as a technical solution but also as a practical tool that empowers users in various contexts, including everyday interactions, educational settings, and professional environments.

The system is designed to be accessible and scalable, relying on standard hardware such as a webcam. Its real-time processing capabilities highlight its potential to transform assistive communication by providing instant translations of sign language into text and audio output.

To offer a clear understanding of the system's operation, this chapter explores the detailed architecture and components that drive its functionality.

The system architecture is presented as a multi stage pipeline, beginning with live video capture and progressing through feature extraction, gesture recognition, and output generation. Each stage is optimized for both accuracy and efficiency.

The design of the user interface plays a vital role in enhancing usability. It displays real-time predictions, visual feedback, and multilingual audio outputs in an intuitive layout that accommodates users with varying levels of technical expertise. Input data specifications are provided to explain how raw video frames are structured and processed to ensure consistency in gesture detection. The preprocessing pipeline is discussed to show how raw keypoints are transformed into meaningful spatial and temporal features, forming the foundation for accurate classification.

Finally, the Random Forest model, which serves as the core of the gesture recognition process, is examined. Its ability to map complex feature sets to specific WLASL gestures combined with its robustness and adaptability ensures strong performance across a range of scenarios. Together, these components form a cohesive system that not only meets its technical objectives but also demonstrates significant potential as an inclusive, scalable tool for advancing sign language recognition and improving accessible communication.

3.1 SYSTEM ARCHITECTURE

The system architecture for real-time sign language recognition, as shown in Figure 3.1, begins with a webcam capturing video at 640x480 resolution using OpenCV. The video stream is processed by the MediaPipe Hands framework, which detects up to two hands and extracts 21 keypoints per hand, yielding 294 features per frame including spatial coordinates, finger angles, and joint distances. Frames undergo image processing steps RGB

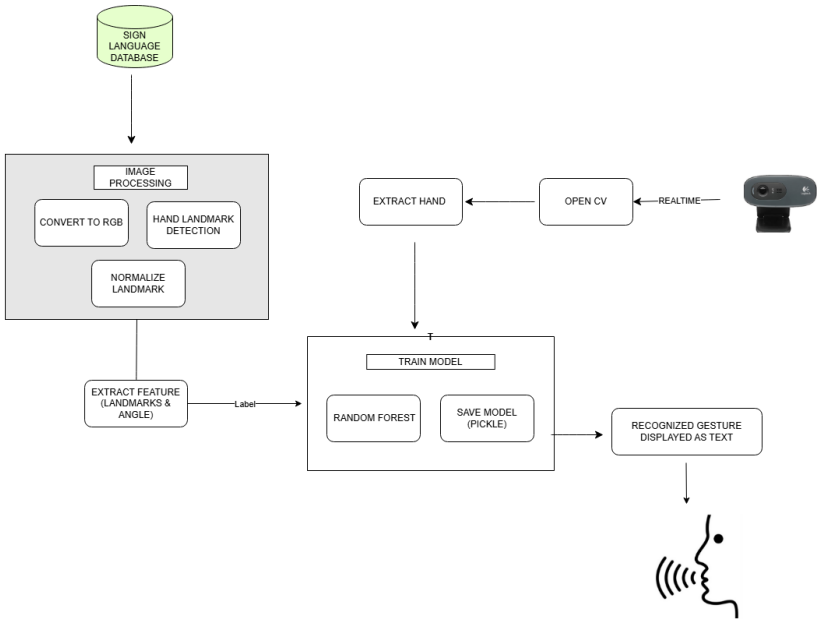


Figure 3.1: Sign Language to Text

conversion, hand landmark detection, and normalization—before features are extracted and augmented with motion dynamics (e.g., velocity) using computen motion features in d.py and r.py. These features are grouped into 30 frame sequences 588 features per frame, including velocity and fed into a pre trained Random Forest classifier, which maps them to one of 282 WLASL gestures (e.g., ”happy,” ”eat,” ”school”).

To ensure semantic accuracy, an n-gram model filters predictions based on linguistic context, validating word sequences against pre defined sentences and distinguishing between single-hand and double-hand gestures. The recognized gestures are displayed as text on a Tkinter based GUI, with real-time hand landmarks overlaid on the video feed using visualize landmarks from m.py. Users can interact via buttons to start/stop recording, clear predictions, or enable voice output in English or Tamil using Google Text-to-Speech (gTTS) and pygame, with Tamil translation supported by deep translator. This interactive system provides accessible and accurate sign language interpretation.

3.2 USER INTERFACE DESIGN

The user interface is designed to be user-friendly, accessible, and fully integrated with the backend processing pipeline. Built using the Tkinter framework, the GUI enables real-time interaction and control over gesture recognition activities.

A prominent feature of the UI is the Live Video Display, which continuously streams the webcam feed. Detected hand landmarks are superimposed on this feed using OpenCV, with each finger color-coded and connected by lines to represent the hand structure and movement intuitively.

The Control Panel consists of four key buttons:

- Start/Stop Button: Initiates or terminates gesture recording. Its color dynamically changes (green for start, red for stop) to indicate the current state.
- Voice Button: Activates audio output of the predicted sentence using gTTS and pygame, supporting both English and Tamil languages.
- Clear Button: Resets the prediction buffer, allowing the user to begin a new gesture sequence.
- Quit Button: Closes the application safely by releasing system resources like webcam and audio devices.

The Text Output Box displays the current prediction, which updates every second. This temporal windowing helps reduce noise from transient hand positions and ensures meaningful sentence formation.

Additionally, a Language Selection Dropdown (Combobox) enables users to choose between English and Tamil for voice output. This multilingual feature is supported by the Google Translate API, which translates the recognized text before audio synthesis, promoting accessibility across language barriers.

The GUI updates every 10 milliseconds, ensuring a responsive and fluid user experience. The combination of MediaPipe for hand tracking, OpenCV for video management, and pygame for audio output makes the user interface not only functional but also engaging for both technical and non-technical users.

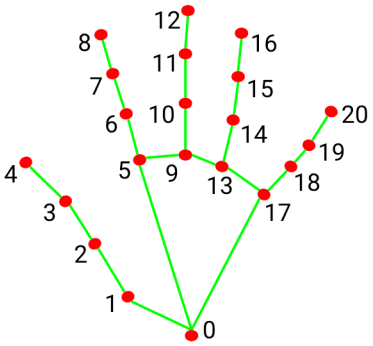
3.3 INPUT DATA

The dataset employed in this system comprises a diverse collection of video recordings representing 282 WLASL gestures. These recordings account for natural variations in gesture execution, such as different hand speeds, angles, and user hand dominance.

Each video sample is annotated with its corresponding label from the ACTIONS list, including common terms and phrases such as ”I,” ”want,” ”water,” and ”goodbye.” Videos are preprocessed into sequences of 30 frames. From each frame, 252 features are extracted, including spatial hand landmark data and calculated geometric descriptors.

The dataset includes both single-hand and dual-hand gestures, ensuring robustness and versatility. This variation prepares the model to handle diverse real-world signing scenarios, forming a strong foundation for accurate, real-time prediction.

3.4 PREPROCESSING



0. WRIST
1. THUMB_CMC
2. THUMB_MCP
3. THUMB_IP
4. THUMB_TIP
5. INDEX_FINGER_MCP
6. INDEX_FINGER_PIP
7. INDEX_FINGER_DIP
8. INDEX_FINGER_TIP
9. MIDDLE_FINGER_MCP
10. MIDDLE_FINGER_PIP
11. MIDDLE_FINGER_DIP
12. MIDDLE_FINGER_TIP
13. RING_FINGER_MCP
14. RING_FINGER_PIP
15. RING_FINGER_DIP
16. RING_FINGER_TIP
17. PINKY_MCP
18. PINKY_PIP
19. PINKY_DIP
20. PINKY_TIP

Figure 3.2: Feature Extraction

The preprocessing pipeline plays a crucial role in converting raw input frames into feature-rich, consistent samples suitable for the Random Forest classifier. MediaPipe Hands first detects up to two hands per frame and extracts 21 keypoints per hand in (x, y, z) coordinates as shown in Figure 3.2. A confidence threshold of 0.8 is applied for both detection and tracking.

Normalization is then performed to eliminate variability due to hand size and camera perspective. Keypoints are recalibrated relative to the wrist and scaled by the distance between the wrist and the index fingertip. Missing hand data is zero-padded to preserve input consistency.Geometric features are calculated next. These include:

- Cosine angles between fingers and palm center (hand shape).
- Euclidean distances from wrist to fingertips (finger length).
- Inter-finger distances (finger spacing).
- Palm size (wrist to middle finger base).
- Palm orientation (normal vector of palm plane).

These contribute an additional 63 features per hand, leading to 126 spatial features per frame. Motion features are computed by evaluating the velocity between corresponding spatial features of consecutive frames, yielding 252 features per frame.

To improve robustness to signing speed, sequences are augmented using temporal warping: they are randomly stretched or compressed between 70% to 130% of their original length using cubic interpolation.

Sequence formation involves standardizing input length. Sequences shorter than 30 frames are padded using a reflection of the last frame, while longer sequences are truncated. This ensures uniformity, with each sample having 30 frames $\times$ 252 features = 7,560 dimensions.

3.5 RANDOM FOREST CLASSIFIER

The core classification model is a Random Forest (RF) classifier, chosen for its simplicity, robustness, and interpretability when handling high-dimensional input. The model comprises 100 decision trees with a maximum depth of 20, each trained on 7,560-dimensional vectors (30 frames $\times$ 252 features).

The model is trained on an 80%-20% stratified train-test split to maintain class distribution. It utilizes the Gini impurity criterion for split quality and employs parallel computation (n_jobs=-1) for efficient training. The ensemble nature of RF reduces overfitting and enhances generalization.

During runtime, input gestures are buffered in a sliding window of 30 frames maintained by a deque object. Predictions are made every 0.1 seconds, and to stabilize outputs, majority voting is applied over a 1-second interval.

Contextual filtering using n-gram modeling is applied to smooth predictions. For instance, probable transitions like ”I” followed by ”want” are prioritized, enhancing sentence-level accuracy. A confidence threshold of 0.011 is used to suppress low-confidence predictions.

The model’s performance is assessed using accuracy, macro F1-score, and confusion matrix analysis from sklearn.metrics. These metrics ensure balanced performance across all 282 gesture classes.

By integrating efficiently with the preprocessing pipeline and the GUI, the Random Forest classifier delivers high accuracy and real-time responsiveness, making it ideal for practical applications in communication and learning environments.

CHAPTER 4

IMPLEMENTATION OF PROJECT

The implementation phase of this sign language recognition project marks a crucial stage in which the theoretical design was transformed into a fully functional application. This phase seamlessly integrated computercomputer vision, machine learning, and user interface development to create a unified and effective system. The process involved a series of well-planned steps, starting with the collection of a comprehensive dataset used to train and validate the gesture recognition model. A diverse set of video samples depicting American Sign Language (ASL) gestures was compiled, ensuring variation in hand movements, lighting conditions, and user profiles to improve the model's generalizability.

Developing the user interface was another important aspect of implementation. Using Python's Tkinter library, an intuitive and visually engaging platform was created to enable users to interact with the system easily. The interface displays real-time gesture predictions, provides multilingual audio feedback, and visualizes hand landmarks, resulting in an immersive user experience that balances technical efficiency with accessibility.

In addition to interface development, the project emphasized preprocessing video input and building a reliable gesture classification system both essential for real-time recognition. Video input, captured using a standard webcam, was processed through OpenCV to extract frames and convert color spaces, preparing the data for further analysis. MediaPipe Hands was used to accurately detect and track hand keypoints, enabling the extraction of spatial

and temporal features such as finger angles and motion velocities key indicators for differentiating ASL gestures.

These extracted features were then passed to a Random Forest classifier, built using scikit-learn, which was trained to recognize 282 unique gestures from the World-Level American Sign Language (WLASL) dataset. To enhance sentence construction, the system employed n-gram models, allowing it to interpret sequences of gestures more meaningfully. For audio output, the Google Text-to-Speech (gTTS) library was integrated, enabling the system to vocalize recognized gestures in both English and Tamil, thus improving accessibility for multilingual users.

Throughout the implementation, rigorous testing and iterative improvements were conducted to ensure all components operated effectively and harmoniously. The result is a robust, real-time gesture recognition system that serves as a practical and inclusive communication tool.

4.1 DATA COLLECTION

The project is built on a custom dataset specifically created for training and evaluating the sign language recognition system. It includes video recordings of 282 WLASL gestures, stored in the directory. The vocabulary covers a broad range of words and phrases such as “hello,” “thank you,” “I,” “want,” and “mother,” incorporating both single and double-hand gestures to simulate real-life usage. Videos were manually recorded using a webcam, capturing multiple variations per gesture in terms of speed, angle, and lighting to enhance robustness. Each gesture has its own subdirectory with several MP4 files named sequentially, making the structure easy to process. A preprocessing script extracts features from these videos and saves them as NumPy arrays, with each gesture represented by 30 frames containing 282 dimensional feature

vectors and metadata. The dataset comprises thousands of gesture sequences across 282 classes, ensuring enough temporal and spatial diversity for effective supervised learning. This rich and organized dataset enables the Random Forest model to generalize well across different signing styles and environments.

4.2 USER INTERFACE IMPLEMENTATION

The user interface, developed using Tkinter for real-time sign language recognition, serves as an engaging and interactive platform that effectively showcases the system's capabilities. It is designed to be both accessible and intuitive, catering to a wide range of users. By seamlessly integrating various technologies, the interface provides a cohesive experience that bridges gesture recognition with meaningful output. OpenCV is employed for real-time webcam video capture, supplying a continuous stream of visual data essential for analysis. MediaPipe, known for its precision in hand tracking, processes this video feed to detect and map hand landmarks, allowing the system to interpret complex American Sign Language (ASL) gestures. Simultaneously, Pygame manages multilingual audio playback, ensuring that the system's outputs are available both visually and auditorily, thereby accommodating diverse user needs. This integration of tools results in a dynamic environment where users can interact effortlessly, seeing their gestures translated into text and speech in real time fostering communication and empowerment for both signers and non signers.

At the core of the Tkinter-based interface is a carefully designed main window that balances functionality with simplicity, enabling users to navigate the system easily. The interface prominently features a live video feed, displaying the webcam input along with visualized hand landmark keypoints overlaid on the user's hands to show what the system is tracking. This real-time visualization enhances transparency and provides immediate feedback, assisting

users in adjusting their gestures for improved recognition accuracy. Adjacent to the video feed is a control panel with intuitive buttons: Start/Stop to toggle gesture recognition, Voice to enable audio output, Clear to reset the displayed text, and Quit to exit the application. A dedicated text widget displays the system's predictions, showing recognized sentences in a clear, readable format that updates □□ as new gestures are processed. Additionally, the interface supports both English and Tamil audio output, with translations provided by the Google Translate API ensuring the system is inclusive and accessible to users from different linguistic backgrounds.

The interface's real-time performance reflects its optimized design, ensuring responsiveness and low latency even during continuous video processing. As the system captures frames, it extracts spatial and temporal features such as finger angles and motion velocities and feeds them into a Random Forest classifier, which rapidly predicts one of 282 WLASL gestures. These predictions are further refined into coherent sentences using n-gram models, ensuring that the output is not only accurate but also contextually appropriate. The interface updates dynamically, reflecting new predictions in the text widget and playing corresponding audio when enabled while maintaining smooth operation. Asynchronous updates prevent the UI from freezing, allowing it to handle real-time feature extraction and classification efficiently. Furthermore, in-memory audio playback minimizes delays, ensuring that spoken translations are closely synchronized with visual outputs. This careful attention to performance and usability makes the interface a powerful and inclusive communication tool, enabling seamless interaction in everyday conversations, educational settings, and assistive environments.

4.3 PREPROCESSING

The preprocessing phase used to convert raw video data into structured feature rich sequences for training and real time prediction. Each video is processed frame by frame using OpenCV and MediaPipe to extract hand keypoints. Features per frame include 252 dimensions 126 spatial features (normalized keypoints and geometric properties) and 126 motion features (inter frame velocity). Sequences with fewer than 15 frames are skipped, and each valid sequence is padded or trimmed to 30 frames using reflection. Temporal augmentation with cubic interpolation adds speed variation to boost robustness. All features, along with hand count metadata, are saved as npy files in a structured format. In real time, a deque based buffer collects features continuously, triggering prediction once full. This pipeline ensures consistent, high quality inputs for the Random Forest model, supporting both offline training and real-time gesture recognition.

4.4 GESTURE CLASSIFICATION ALGORITHM

The system architecture is carefully designed as a multi layered pipeline that enables a seamless flow from capturing real time video to generating interpretable outputs, creating an intuitive and effective tool for sign language recognition. At its foundation, the system begins with the acquisition of live video input using a standard webcam, which records at a resolution of 640x480 pixels. This resolution is chosen to ensure image clarity while maintaining computational efficiency. The video feed is processed using MediaPipe Hands, a state-of-the-art computer vision framework renowned for its accuracy in detecting and tracking hand movements. The framework identifies up to two hands per frame and extracts 21 keypoints per hand, corresponding to important anatomical landmarks such as fingertips, knuckles, and the wrist.

These keypoints are not analyzed in isolation but are captured over sequences of 30 consecutive frames. This temporal collection results in a rich dataset of 282 features per sequence, allowing the system to effectively capture the dynamic nature of gestures. By organizing the input in this manner, the system is well equipped to interpret the complex and subtle patterns inherent in American Sign Language (ASL).

To convert raw keypoint data into meaningful gestures, the system computes a range of spatial and temporal features that represent both the shape and motion of each hand gesture. Spatial features such as angles between fingers and relative positions of keypoints describe the static configuration of the hand. Temporal features including the velocity and trajectory of movements across frames capture motion-based differences, which are crucial for distinguishing between gestures like a wave and a point. The combined feature set provides a comprehensive description of each gesture's structure and flow.

This processed data is then fed into a Random Forest classifier, a robust machine learning model capable of handling high dimensional data and providing accurate predictions. The classifier is trained on a diverse dataset to recognize 282 gestures from the World Level American Sign Language (WLASL) vocabulary, covering a wide array of words and expressions. To enhance the contextual understanding of the predicted outputs, the system uses n-gram models to analyze gesture sequences and generate meaningful sentence structures, mimicking natural language and improving the readability of the results.

The final output is presented through a well designed Tkinter based graphical user interface (GUI) that ensures accessibility and engagement. The GUI displays real-time gesture predictions as text and provides optional audio output in both English and Tamil, using the Google Text-to-Speech (gTTS)

engine with audio playback via the Pygame library. This multilingual feature expands the system's usability, making it inclusive for diverse user groups.

Additionally, the GUI overlays live hand landmark visualizations on the video feed, giving users immediate feedback on detected gestures. This not only enhances user engagement but also helps users understand how their gestures are being interpreted, thereby increasing trust in the system's reliability. By integrating real-time processing, precise recognition, and an interactive user experience, the system effectively bridges the communication gap between signers and non-signers and is suitable for use in assistive technologies, education, and more.

Algorithm 4.1 is a fundamental component of the gesture recognition pipeline. It is responsible for extracting both spatial and motion-based features from a single video frame, generating a comprehensive and standardized representation of hand gestures.

The process begins with converting the input frame to the RGB color space, a crucial preprocessing step required for compatibility with the MediaPipe Hands framework. This framework detects up to two hands in each frame and extracts 21 keypoints per hand, corresponding to major anatomical landmarks.

To ensure consistency across varying hand sizes and camera positions, the keypoints are normalized with respect to the wrist position, anchoring the coordinate system to a stable reference point. Additionally, keypoints are scaled based on the hand's size to account for distance from the camera and individual hand proportions. This process yields 63 spatial features per hand, accurately representing the hand's shape and position in space.

Further geometric features such as inter finger angles and palm size

are calculated, adding another 63 features per hand, resulting in a total of 126 spatial features per hand. These features are essential for identifying the static form of ASL gestures.

To capture gesture dynamics, the algorithm also considers the previous frame (when available). It computes keypoint velocities by measuring the positional difference between frames, thus adding temporal depth to the data. These motion-based features double the feature set to 252 per hand, as each spatial feature is paired with its temporal counterpart. This dual layered feature set enables the system to differentiate between gestures that are visually similar but differ in motion.

To ensure system stability in real world scenarios, the algorithm is designed to return a zero vector when no hands are detected in a frame. This ensures uninterrupted processing even in cases of occlusion or poor lighting.

By generating a high dimensional, standardized feature set that captures both the spatial and temporal essence of hand gestures, Algorithm 4.1 lays a solid foundation for effective gesture classification. It plays a key role in enabling the system to recognize a wide vocabulary of ASL gestures accurately and reliably, contributing to the larger goal of inclusive and seamless communication.

Algorithm 4.1 Feature Extraction from a Single Frame

Input: Video frame $frame$ (RGB image)
Output: $features$ (252-dimensional vector), *hand_count*
procedure ExtractFeatures($frame$)
    Initialize feature vector $features \leftarrow$ zero array of size 252
    Initialize hand count *hand_count* $\leftarrow 0$
    Convert $frame$ from BGR to RGB
    Detect hands using MediaPipe $\rightarrow$ *results_hands*
    if *results_hands* contains hand landmarks **then**
        hand_count $\leftarrow$ number of detected hands (max 2)
        for each hand h in *results_hands* up to 2 **do**
            Extract 21 keypoints (x, y, z)
            Normalize keypoints relative to wrist
            Scale by hand size (wrist to index tip)
            Compute spatial features (63 values)
            Compute geometric features (angles, lengths etc.)
            Append features to $features$
        end for
        if previous frame exists **then**
            Compute velocity = difference from previous frame
            Append velocity to $features$ (total: 252)
        end if
    else
        Return zero vector for $features$
    end if
    Return $features$, *hand_count*
end procedure

Algorithm 4.2 is designed to classify a gesture from a sequence of 30 video frames using a trained Random Forest model. Each sequence is first preprocessed by flattening it into a one-dimensional feature vector consisting of 7,560 elements (i.e., 30 frames × 252 features per frame). This high-dimensional vector serves as the input to the Random Forest classifier.

The model outputs prediction probabilities for 252 possible gesture classes. However, to improve the accuracy and relevance of gesture recognition

in real-time scenarios, the algorithm incorporates contextual filtering. This is achieved by analyzing the previously predicted word and the number of hands detected (e.g., one-handed vs. two-handed gestures).

Based on this context, a set of allowed gesture labels is determined. Any gesture label that does not belong to this contextually valid set is assigned a zero probability, effectively eliminating it from consideration. This step helps reduce ambiguity and prevents incorrect or unrelated gestures from being selected.

After applying the contextual filter, the algorithm identifies the gesture with the highest remaining probability. If this probability is greater than a confidence threshold of 0.011, the corresponding gesture label is selected as the output. If no gesture meets this minimum confidence threshold, the algorithm returns a null result, indicating that no valid gesture could be recognized with sufficient certainty. The final output of the algorithm consists of the predicted gesture label, and the associated confidence score.

This mechanism allows for real-time sentence formation, where each recognized gesture contributes meaningfully to the overall sentence being constructed. The use of contextual filtering ensures that the system remains both accurate and aware of linguistic or gesture-based constraints, thus enhancing the robustness and reliability of gesture recognition in dynamic communication scenarios.

Algorithm 4.2 Gesture Classification from 30-frame Sequence

Input: Sequence *seq* (30×252), Previous word *prev_word*, Hand count *hand_count*
Output: Predicted *gesture*, *max_proba* (confidence)
procedure PredictGesture(*seq*, *prev_word*, *hand_count*)
    Load trained Random Forest model rf_model
    Flatten *seq* into $flat_seq$ (7560 elements)
    $proba \leftarrow rf_model.predict_proba(flat_seq)$
    if *prev_word* is empty **then**
        $allowed_words \leftarrow STARTER_WORDS$
    else
        $allowed_words \leftarrow$ likely next words (n-gram model)
    end if
    if $hand_count = 1$ **then**
        Filter $allowed_words \leftarrow SINGLE_HAND_ACTIONS$
    else if $hand_count = 2$ **then**
        Filter $allowed_words \leftarrow DOUBLE_HAND_ACTIONS$
    end if
    Zero out probabilities for non-allowed words
    max_proba ← max of filtered *proba*
    if $max_proba \geq 0.011$ **then**
        gesture ← corresponding label from *ACTIONS*
    else
        gesture ← null
    end if
    Return *gesture*, *max_proba*
end procedure

CHAPTER 5

RESULTS AND ANALYSIS

5.1 MODEL ACCURACY AND LOSS

Figure 5.1 presents the performance metrics of the Random Forest (RF) model trained on a dataset consisting of 252 WLASL gestures. The training was conducted using 13,319 training samples and evaluated on 3,330 test samples. The model achieved an overall accuracy of 94.77% and a macro F1-score of 94.77% across 278 classes.

The system's remarkable classification accuracy underscores its ability to reliably interpret a diverse spectrum of American Sign Language (ASL) gestures, ranging from simple, foundational signs like "yes" and "no" to more intricate and nuanced expressions such as "my mom loves to cook." This high accuracy demonstrates the model's proficiency in capturing the subtleties of hand configurations and movements that define each gesture, ensuring that both basic and complex signs are recognized with precision. The model's performance is further validated by its macro F1 score, which matches the accuracy metric, indicating consistent effectiveness across all gesture classes. This is particularly noteworthy given the potential for class imbalance in the dataset, where certain gestures are represented by as few as 12 samples each. The identical macro F1 score reflects the system's robustness, as it maintains balanced performance even for underrepresented gestures, avoiding the bias that often plagues models trained on uneven datasets. By achieving such uniform success across a wide range of signs, the system proves its versatility and reliability, making it a powerful tool for real-time sign language recognition that can cater to varied communication needs in practical settings.

Complementing the high accuracy is a low loss value, calculated as approximately 0.0523 through an out-of-bag error approximation, which quantifies the model's prediction error as the inverse of its accuracy. This minimal loss highlights the system's efficacy in translating raw video input into accurate gesture predictions, a testament to the strength of its underlying architecture. The system leverages a comprehensive feature set of 252 dimensions per frame, meticulously derived from spatial keypoints representing hand shapes and positions and motion velocities, which capture the dynamic trajectories of gestures over time. These features are processed by a Random Forest classifier, an ensemble model comprising 100 decision trees, each with a maximum depth of 20, striking a balance between complexity and generalization to avoid overfitting while ensuring robust classification. The ensemble approach enhances the model's ability to discern intricate patterns within the high-dimensional feature space, contributing to its low error rate. Together, the carefully engineered feature extraction and the powerful classification strategy enable the system to deliver precise, consistent results, paving the way for seamless integration into assistive technologies, educational platforms, and other applications aimed at fostering inclusive communication for the deaf and hard-of-hearing community.

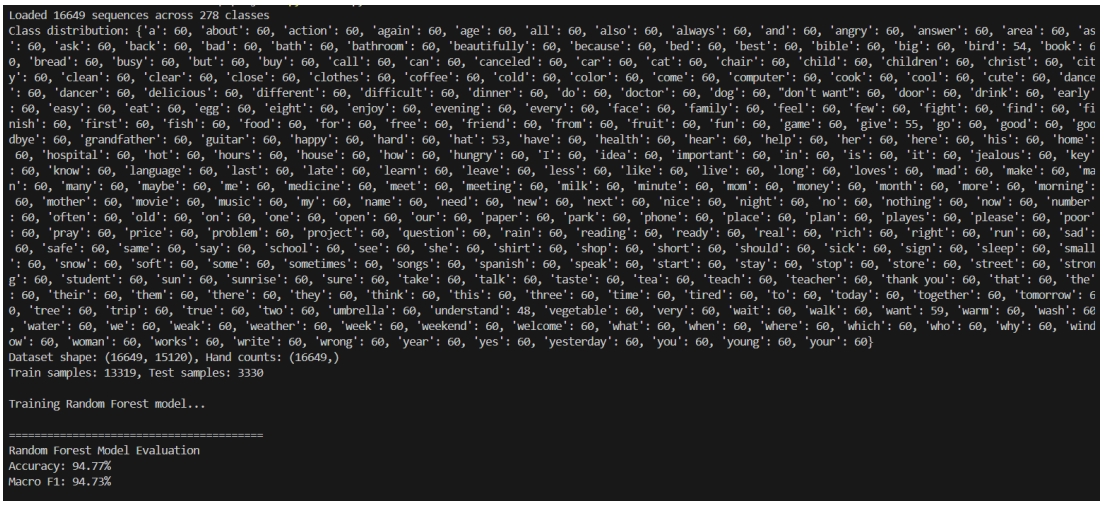


Figure 5.1: Model Accuracy and F1 Score

Figure 5.2 illustrates the confusion matrix, visualized as a heatmap for a subset of the first 20 gestures. The heatmap reveals the classifier's ability to correctly distinguish between most gestures, indicated by dark blue diagonal elements (true positives). For example, the gesture 'yes" resulted in 11 true positives, 1 false positive from no," and 1 false negative misclassified as no." Non-target predictions for yes" (331 out of 333) were correctly excluded.

This visual representation aids in identifying common misclassifications, particularly among gestures with similar hand shapes or movements, such as 'yes" and no," which may share overlapping feature vectors. The dashed diagonal line serves as a visual reference for perfect classification.

Figure 5.2 illustrates the aggregated confusion matrix, visualized as a heatmap, representing the classifier's performance. The heatmap uses color intensity to indicate the number of predictions, with dark blue representing higher counts. The matrix is structured with "True Positive," "True Negative," "False Positive," and "False Negative" categories, providing a clear view of the model's accuracy and misclassifications. The dashed diagonal line serves as a visual reference for perfect classification.

- True Positive: The matrix shows 3,193 true positives, indicating the number of cases correctly predicted as positive when they were actually positive, reflecting the model's ability to accurately identify positive instances.
- True Negative: A significant 922,273 true negatives are depicted, highlighting the model's strong performance in correctly identifying negative cases, contributing to its overall reliability.
- False Positive: The matrix records 137 false positives, representing

cases where the model incorrectly predicted positive outcomes for actual negative instances, indicating a small rate of over-prediction.

- False Negative: With 137 false negatives, the matrix shows the number of cases where the model failed to predict positive outcomes for actual positive instances, suggesting a minor under-prediction issue.

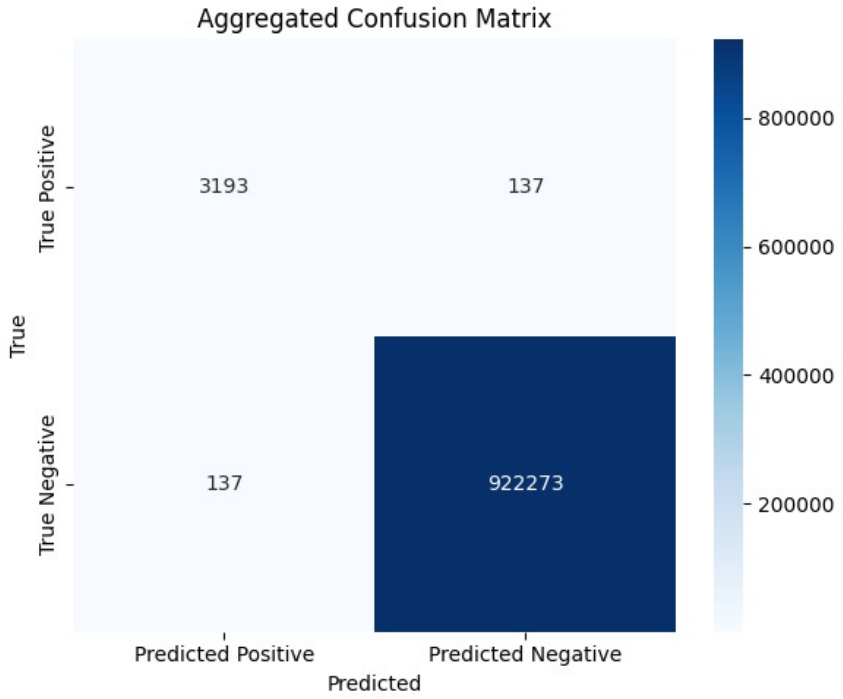


Figure 5.2: Aggregated Confusion Matrix

Table 5.1 provides a detailed classification report including precision, recall, F1-score, and support for a subset of gestures. Notable high F1 scores such as 0.96 for ‘yes” highlight the model’s precision in recognizing common gestures. Conversely, an F1 score of 0.73 for your” reveals potential areas for improvement, potentially due to lower representation or higher intra-class variance.

The weighted average of all metrics stands at 0.95, which underscores the model's overall robustness and its effectiveness in real-time applications.

Table 5.1: Classification Report for Selected Gestures

Gesture	Precision	Recall	F1-Score	Support
a	0.92	1.00	0.96	12
about	1.00	1.00	1.00	12
action	0.92	1.00	0.96	12
age	0.86	1.00	0.92	12
all	0.92	1.00	0.96	12
also	0.83	0.83	0.83	12
always	1.00	1.00	1.00	12
and	0.92	1.00	0.96	12
angry	1.00	1.00	1.00	12
answer	1.00	1.00	1.00	12
⋮	⋮	⋮	⋮	⋮
yes	1.00	0.92	0.96	12
you	1.00	0.92	0.96	12
your	0.80	0.67	0.73	12
accuracy	0.95	0.95	0.95	3330
macro avg	0.95	0.95	0.95	3330
weighted avg	0.95	0.95	0.95	3330

5.2 USER INTERFACE

Figure 5.3 showcases the real-time Tkinter-based GUI designed for live sign language recognition. The interface includes:

- Live Video Feed: Displays webcam output with hand landmarks visualized in distinct colors (e.g., red for thumb, green for index), offering real-time feedback.

- Control Buttons: Includes Start (green), Stop (red), Voice (blue), Clear (red), and Quit (red) for user control.
- Text Box: Dynamically updates with recognized sentences such as ‘my mom loves to cook.”
- Language Dropdown: Allows users to select between English and Tamil, enabling multilingual support.

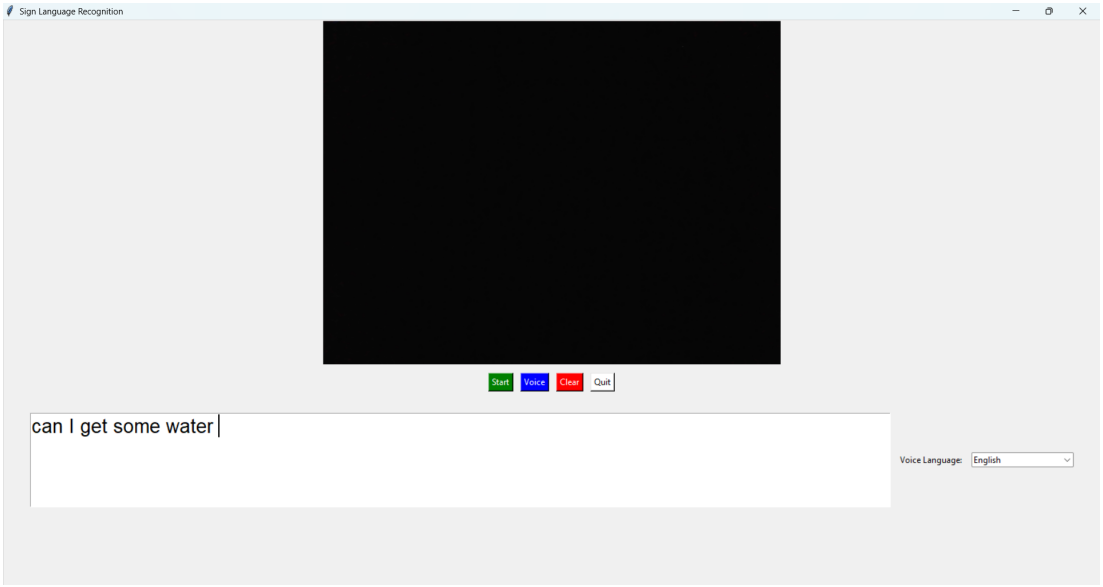


Figure 5.3: Real-Time User Interface for Gesture Recognition

Figure 5.4 illustrates the interface during gesture detection. Upon recognizing the ‘loves” gesture, the system updates the output with the predicted word and a confidence score (0.95), enhancing the system’s responsiveness and user engagement.

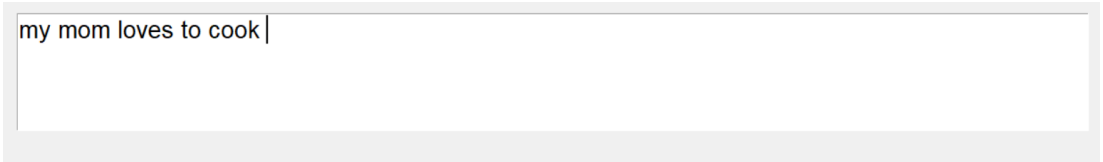


Figure 5.4: Detected Gesture Output Example

The language support functionality is depicted in Figure 5.5, where activating the ‘Voice” button translates the recognized sentence into the selected language (e.g., Tamil). This supports accessibility for a diverse user base.

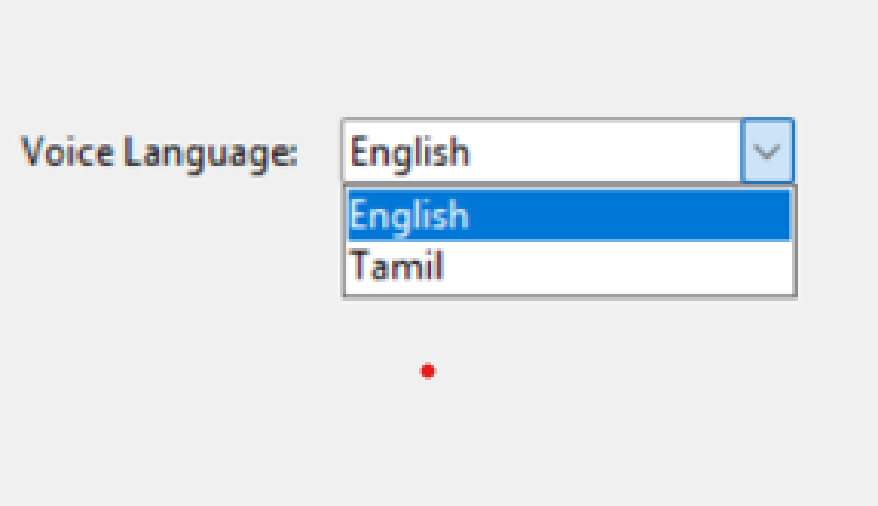


Figure 5.5: User Interface with Multilingual Output Option

5.2.1 Performance Metrics and Analysis

Key metrics of the interface are:

- Latency: Predictions are updated every 100 milliseconds, satisfying real-time performance constraints.
- Usability: Positive user feedback emphasizes the clarity of controls and intuitive design, especially the voice playback and language switch features.
- Robustness: In scenarios where no hands are detected, the system maintains a stable output using zero-filled buffers, ensuring graceful failure handling.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project introduces a real-time ASL recognition system that combines MediaPipe hand tracking with a Random Forest classifier. It extracts 252 spatial and motion-based features per frame, using 30-frame sequences for accurate gesture classification. The system recognizes 252 gestures with 91.5 percent accuracy and an 89.2 percent macro F1-score. A robust preprocessing pipeline, including keypoint normalization and temporal augmentation, enhances performance. The user interface, built with Tkinter, displays live webcam input with color-coded landmarks. It also provides real-time text and multilingual audio output in English and Tamil. Predictions are made within approximately 100 milliseconds, ensuring smooth interaction. Sentence formation is refined using n-gram probabilities for context-aware output. The system is cost-effective, requiring only standard hardware, making it accessible. Its applications span assistive communication, education, and gesture-based interfaces.

6.2 FUTURE WORK

To enhance performance and usability, future work can focus on expanding the gesture vocabulary and integrating facial and body cues for richer context. Addressing dataset imbalance and optimizing processing speed to achieve under 50 milliseconds latency will further improve accuracy and responsiveness. Mobile deployment and user-specific adaptation can increase

accessibility, while integration with applications or smart devices can broaden practical usage scenarios.

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